Problem Set 2

Epidemiology and Population Health

Jay Ballesteros

October 13, 2026

Section time: 2:30-3:20 PM

Q1.

This is drawn from an investigation of an outbreak of Staphylococcal food poisoning following a wedding ceremony in rural India in February 2018. The outbreak was investigated by the Epidemic Intelligence Service Office under the National Centre for Disease Control in India. They interviewed 59 attendees who experienced vomiting or diarrhea within 24 hours after the wedding and another 59 attendees or neighbors from the same village who were not ill.

Below are the data derived from the interviews about which foods or drinks the attendees ate at the wedding.

Food	ATE: Ill	ATE: Not ill	ATE: Total	ATE: Attack rate	DID NOT EAT: Ill	DID NOT EAT: Not ill	DID NOT EAT: Total	DID NOT EAT: Attack rate	Risk ratio*
Gaajar halwa	56	35	91	62%	3	24	27	11%	5.54
Paneer	45	40	85	53%	14	19	33	42%	1.25
Dal rice	58	53	111	52%	1	6	7	14%	3.66
Water	43	50	93	46%	16	9	25	64%	0.72
Papad	34	40	74	46%	25	19	44	57%	0.81
Pickle	17	20	37	46%	42	39	81	52%	0.89

Food	ATE: Ill	ATE: Not ill	ATE: Total	ATE: Attack rate	DID NOT EAT: Ill	DID NOT EAT: Not ill	DID NOT EAT: Total	DID NOT EAT: Attack rate	Risk ratio*
Okra fry	30	38	68	44%	29	21	50	58%	0.76
Green salad	28	41	69	41%	31	18	49	63%	0.64
Kothimbir Vadi	24	39	63	38%	35	20	55	64%	0.59
Poori aloo	19	36	55	35%	40	23	63	63%	0.54
Pineapple Juice	8	20	28	29%	51	39	90	57%	0.50
Cabbage	4	11	15	27%	55	48	103	53%	0.50

*Risk ratio is the ratio of the attack rates in those who ATE the food over those who DID NOT EAT the food.

a. Determine the attack rates and risk ratios among those who ate the specified items and complete the table. (18 points, 0.5 for each attack rate and risk ratio)

Answers above!

b. Which item shows the highest attack rate? Is the attack rate low among persons not exposed to that item? (6 points)

The highest rate is for people that ate *Gaajar halwa* (62%). And yes! Way more lower (11%). The risk ratio is 5.54.

c. Write a sentence that interprets the risk ratio for the item most likely to be responsible for the outbreak. (3 points)

The risk ratio of 5.54 for *Gaajar halwa* indicates that individuals who consumed this food (exposure) had 5.5 times the risk of illness compared to those who did not consume it. This suggests a strong association between eating *Gaajar halwa* and the outbreak of illness.

d. What is the odds of illness for people exposed to that food item? (3 points)

$$\frac{\text{ATE: ill}}{\text{ATE: not ill}} = \frac{56}{35} = 1.6$$

e. What is one possible reason why there were cases among those who did not eat this food? (3 points)

Maybe cross-contamination having spill-over effects. It's possible that, while handling food, utensils came into contact with several dishes. For example, serving spoons.

Additionally, recall bias by some people during interviews. For instance, maybe someone had a bite of Gaajar halwa and forgot, or simply decided not to report it.

f. If some of the foods investigated turned out to be protective against becoming ill, what would be the range of their risk ratios? What would be one reason why a food would be found to be protective? (6 points)

A protective food would have a risk ratio between 0 and 1 ($0 \leq$ risk ratio < 1). A risk ratio of 1 means no association, and one less than 1 means a lower risk among those who ate the food. Finally, a risk ratio of 0 means that no person who ate the food became ill.

One reason why a food can look protective is confounding. For instance, people who ate more of the foods with a risk ratio < 1 possibly ate less Gaajar halwa (or none). Therefore, the "protective" effect possibly comes from less exposure to the contaminated food.

g. The data were also stratified by gender, and the risk ratio was much higher for the women who ate the implicated food than for the men who ate it. In most of the community gatherings in rural India, women usually eat after men. Given this information, why might the risk be higher for women than men? (3 points)

Staphylococcal food poisoning gets worse when contaminated food stays at room temperature, because the bacteria multiply and produce more toxin[1]. In rural India, women usually eat after men, so the Gaajar halwa that women ate had stayed at room temperature for a longer time. Thus, women likely ingested a larger dose of toxin than men, which increased their risk of becoming ill.

[1] https://www.msdmanuals.com/professional/gastrointestinal-disorders/gastroenteritis/staphylococcal-food-poisoning